Umair Javaid Manj

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Research Experience

Doctoral Researcher — **Auckland University of Technology**, Health & Artificial Intelligence, Auckland — Oct 2025 to Present

- Developing an AI based biomarker of chronic low back pain from cortical responses to paraspinal stimulation.
- Extending fMRI mapping of the back (Cole et al., 2022; Guekos et al., 2023; Dörig et al., 2026) into the EEG dimension, adding millisecond timing and oscillatory dynamics, and comparing subluxated with non subluxated spinal segments within the same person.
- Designing a case control EEG study combining source imaging, representational similarity analysis and subject level machine learning.
- Led the first pilot and identified an analysis artefact behind an apparent decoding effect.

Data & AI Governance Analyst — **New Zealand College of Chiropractic**, Auckland — Aug 2026 to Present

- Analyse EEG hyperscanning of chiropractor and patient, estimating inter brain and within brain connectivity (wPLI, dPLI) and source level networks.
- Apply graph theory to resting state EEG networks in a randomised chiropractic trial of adults and children, with linear mixed models.
- Model longitudinal clinical outcomes (immune response, DASS-21, PROMIS-25) with mixed effects models in R.
- Manage private participant data and consent records.

Program Manager — **National University of Sciences and Technology**, Islamabad — Sep 2024 to Oct 2025

- Led a national AI training program.

Instructor — **AI Lounge**, Islamabad — May 2023 to Jul 2024

- Supervised 18 student AI projects; awarded Best Instructor.

Research Associate — **COGNET Lab, SEECS**, National University of Sciences and Technology, Islamabad — Aug 2021 to Apr 2023

- Researched the robustness of deep learning and computer vision models to adversarial inputs.
- Mentored junior research assistants (Research Assistant until Nov 2022).

Publications

- Akhter, M. N., Ali, H., Niazi, I. K., **Manj, U. J.**, & Taylor, D. (2026). Application of machine learning techniques in the measurement of ground reaction force (GRF) waveforms using inertial measurement unit (IMU) sensors: a systematic literature review. *Medical Engineering & Physics*. doi
- Rizvi, D. Q., Majeed, M., **Manj, U. J.**, & Haq, M. I. U. (2025). Sleep disorder prediction with PSO-optimized machine and deep learning models. *ICODT2*, pp. 1 to 6. doi
- Haq, M. I. U., Ahmad, Z., **Manj, U. J.**, Akhter, M. N., Arfeen, Z. A., & Qamar, S. D. (2026). Bio-inspired soft robotic gripper with variable stiffness using magneto-rheological control. *ICRAI*, pp. 1 to 6. doi
- Haq, M. I. U., **Manj, U. J.**, Paracha, S. Q., Imam, Y., Rizvi, D. Q., & Farooqi, A. R. (2026). Neuro-symbolic AI-driven secure communication framework for autonomous cyber-physical systems. *APAN 2026*, pp. 1 to 6. doi
- Akhter, M. N., **Manj, U. J.**, Ali, H., Niazi, I. K., & Taylor, D. Deep learning-based kinetics and center of pressure estimation during sit-to-walk transitions from a single sacrum-mounted IMU. *IEEE Transactions on Biomedical Engineering* (under review).

- Akhter, M. N., Ali, H., Niazi, I. K., **Manj, U. J.**, & Taylor, D. Predicting ground reaction force during sit to walk: towards kinetically instrumenting the Timed Up and Go test. *Medical & Biological Engineering & Computing* (under review).
- **Manj, U. J.**, Rizvi, S. D. Q., Niazi, I. K., Akhter, M. N., & Ali, H. Gated feature fusion and age confounding in plantar thermography for diabetes screening. *Biomedical Physics & Engineering Express* (under review).
- **Manj, U. J.**, Qamar, D., Rashid, M. A., Perwaiz, N., & Niazi, I. K. Speaker leakage and subject level accuracy in voice biomarkers for Parkinson's disease. *Machine Learning: Health* (under review).

Education

PhD	**Auckland University of Technology**, Health & Artificial Intelligence, Auckland, New Zealand	Oct 2025 to Present
MS	**National University of Sciences and Technology**, Computational Science & Engineering, Islamabad	2017 to 2021
BE	**National University of Sciences and Technology**, Computer Engineering, Islamabad	2012 to 2016

Skills

Neurophysiology & Signals EEG preprocessing (ICA, ASR, GEDAI), ERP and time frequency analysis, source localisation, functional connectivity (wPLI, dPLI) and graph theory, hyperscanning, IMU analysis, vibrotactile stimulation

EEG Toolboxes MNE-Python, autoreject, EEGLAB (ICLabel, ROIconnect), FieldTrip, BrainNet Viewer

Machine Learning CNN transfer learning (DenseNet, ResNet, ConvNeXt), multi backbone feature fusion with gating and squeeze and excitation, TabNet, TabPFN, transformers and graph neural networks; gradient boosting (XGBoost, LightGBM, CatBoost), SVM, random forests, regularised logistic regression, stacking ensembles

Model Evaluation nested and subject grouped cross validation, leave one subject out, leakage diagnostics, confounder control, Bayesian optimisation (Optuna, Gaussian processes) and particle swarm optimisation, calibration (ECE, Brier), explainability (SHAP, LIME)

Statistics hypothesis testing (exact McNemar, Wilcoxon signed rank) with Holm correction, cluster bootstrap and Clopper-Pearson confidence intervals, covariate adjusted AUC, permutation tests, linear mixed models (lme4, lmerTest, emmeans), cluster based permutation for EEG, power analysis (G*Power)

Programming Python (PyTorch, timm, scikit-learn, NumPy, SciPy, pandas, NetworkX), R, MATLAB, C++; HPC with Slurm

Research Practice Participant data governance, systematic review, scientific writing